MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MICROWAVE (NEGATIVE GRID)

TYPE 7698

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, planar, ceramic-metal
See figure 1
Mounting position: Any
Weight: 1.8 ounces (48 grams) nominal

ABSOLUTE RATINGS:

Parameter:	F	Ef	Eb	epy	Ec	ib	Ib	ic	tp	Du
Unit:	GHz	V	Vdc	kv	Vdc	a	mAdc	a	μs	---
Maximum:										
Anode pulsed osc or amp:	3.0	6.3 ±5% Notes 1 and 12	---	3.5	-150	5 Note 2	---	2.5	3.5	0.0025
Grid pulsed osc or amp:	3.0	6.3 ±5% Notes 1 and 12	2,000	---	-150	5 Note 2	---	2.5	3.5	0.0025
TEST CONDITIONS:	---	6.3	600	---	Adj	---	25	---	---	---

ABSOLUTE RATINGS:

Parameter:	Pp	Pg	tk	TE	T Anode shank	Barometric pressure, reduced	Cooling
Unit:	W	W	sec (min)	°C	°C	mmHg	---
Maximum:							
Anode pulsed osc or amp:	10	2 Note 3	60	250 Note 4	250 Note 4	35 Note 5	Note 4
Grid pulsed osc or amp:	10	2 Note 3	60	250 Note 4	250 Note 4	35 Note 5	Note 4
TEST CONDITIONS:	---	---	300	---	---	---	Note 6

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	7							
1211	Insulation of electrodes	-	Eb = Ek = 0; Ec = -500 Vdc	0.65	II	R	50	---	Meg
1236	Power oscillation (pulse)	8,9	F = 3.0 GHz(min); epy = 3.5 kv; Ec = -1.5 V(min); Rg/Ib = 12 mAdc(max); Ef = 5.8 V	0.65	II	Po	6.0	---	W
1261	Electrode voltage (1) (grid)	6		0.65	II	Ec	-4.5	-9.5	Vdc
1266	Total grid current	6		0.65	II	Ic	---	-2.0	µAdc
1231	Pulsing emission	-	eb = ec = etd/ is = 10a; tp = 3 µs (max); prr = 600 (max)	0.65	II	etd	---	180	v
1301	Heater current	-		0.65	II	If	1.20	1.40	A
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance	9	No voltages applied; use fixture per Drawing 158-JAN	---	---	Cgk Cpg Cpk	7.00 2.10 ---	9.00 2.40 0.06	pF pF pF
---	Resonance	9,11	No voltages applied	---	---	---	---	---	---
1261	Electrode voltage (2) (grid)	-	Ec/Ib = 1.0 mAdc; Eb = 1,000 Vdc	---	---	Eco	---	-30	Vdc
(C)---	Power gain	13	F = 1,100 +50 MHz; Ebb = 2,200 Vdc (min); Ecc = -50 Vdc; tp = 3 µs (min); Du = 0.002 (min); pd = 400 w	---	---	po	1.8	---	kw
	Quality conformance inspection, part 3								
(C)---	Life test	10	Group C; t = 500 hours; filament standby Ef = 6.3 Vac	---	---	---	---	---	---
---	Life-test end point	10		---	---	Δib	---	25	%
---	Variable-frequency vibration	9,14, 15	55 to 500 Hz at 10 G peak; Ec/Ib = 10 mAdc; Rp = 10,000 ohms; Ebb = 300 Vdc	---	---	Ep	---	250	mVac
---	Torque	9,15, 16	No voltages applied	---	---	---	---	---	---
---	Shock	9,15, 17	60 G peak (min); 11 ±2 ms; no voltages applied	---	---	---	---	---	---
---	Torque and shock test end point:								
1266	Total grid current	-		---	---	Ic	---	-10	µAdc
(C)1002	Barometric pressure, reduced	15,18	Pressure = 35 mmHg (max); voltage = 2,000 Vac; T_A = 30° ±10°C						

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NOTES:

1. The transit time heating effect of the cathode shall be compensated by a reduction in heater voltage after dynamic operation of the tubes has started. The back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for a particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. No reduction in heater voltage is required up to and including 500 MHz.

2. The regulation or series-anode-supply impedance, or both, shall limit the instantaneous peak current, with the tube considered as a short circuit, to a maximum of 10 times the specified maximum current rating.

3. The maximum instantaneous peak grid-cathode voltage shall be within the range of +250 to -750 volts.

4. Sufficient conduction and convection cooling shall be provided to limit the envelope and anode shank temperatures to the specified maximum value under all operating conditions. Reliability will be seriously impaired if this maximum is exceeded. Where emphasis is placed on long and reliable life, lower temperatures shall be used.

5. Operation at this altitude is possible in a suitably designed circuit.

6. Sufficient conduction, convection and forced-air cooling shall be used in all electrical tests involving application of heater voltage to maintain the anode shank and seal temperatures within the specified maximum values.

7. All tests listed under quality conformance inspection, part 1, shall be performed at the conclusion of the holding period.

8. The applied voltage pulse shape shall be measured with a noninductive resistor of 1,150 ohms ±2 percent inserted in place of the tube. The pulse shape shall be tp = 3.0 µs ± 10 percent, tr = 0.4 µs (max), tf = 0.7 µs (max). The pulse repetition rate shall be adjusted so that Du = 0.0025 ± 5 percent with the above measured pulse length. Test in cavity in accordance with Drawing 279-JAN. The cavity shall be connected to a load with a VSWR less than 1.5. The oscillator output coupling and the grid or cathode resistor may be adjusted for maximum power output.

9. Other tube contact configuration may be used provided the tube contact area remains unchanged and the socket, jig, or cavity gives equal performance. Mounting of the socket, jig, or cavity may be at the option of the manufacturer.

(C) 10. At zero hours, establish the drive conditions necessary to obtain 3.0 amperes peak anode current with an anode voltage of 1,000 Vdc and a bias voltage of -40 Vdc. The pulse width of the modulator shall be 2 µs (minimum) and the duty shall be 0.0025 maximum. With the drive level determined at zero hours, check the anode current at the end of life. The maximum allowable drop in anode current is 25 percent.

11. Grid-anode resonance. Test in cavity in accordance with Drawing 278-JAN. Cavity shall resonate at 1,354 ± 2.0 MHz with tuning slug in accordance with Drawing 277-JAN at TA = 25° ±5°C.

Grid-cathode resonance. Test in cavity in accordance with Drawing 283-JAN. Cavity shall resonate at 1,719 ± 2 MHz with tuning slug in accordance with Drawing 277-JAN at TA = 25° ± 5°C. When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode capacitance, the tube under test shall be represented by a point within a parallelogram whose four corners are located by the following points:

Points	Capacitance (pF)		Frequency (MHz)	
	C-gp	C-gk	F-gp	F-gk
1	2.10	7.0	1,935	1,765
2	2.10	7.0	1,965	1,795
3	2.40	9.0	1,870	1,725
4	2.40	9.0	1,900	1,755

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NOTES: -Continued

12. Where emphasis is placed on long and reliable life, the filament voltage can safely be lowered to 6.0 volts, provided the line voltage is regulated better than ±2 percent.

13. Test shall be conducted in power amplifier cavity as shown on figure 2. Driving power is defined as the net power delivered to the amplifier input terminals and the reflected power shall be subtracted from the incident power to obtain the net driving power. The output tuning shall be adjusted for maximum power output.

14. The tube shall be mounted in socket in accordance with Drawing 276-JAN and vibrated with simple harmonic motion. The peak acceleration over the frequency range shall be within ±20 percent of the reference acceleration at 100 Hz. The frequency shall vary from 55 to 500 Hz and return to 55 Hz with approximately logarithmic progression and shall require 4 minutes minimum, 6 minutes maximum, to traverse the range. Each tube shall be vibrated for 30 minutes in each position X and Z except that if the cumulative result of test on 50 or more tubes of a construction show that more than 75 percent of the tubes have higher output voltages in one position, subsequent measurements need to be taken only in the position giving the higher reading. The voltages specified herein shall be applied to the tube during vibration. The value of the alternating voltage, Ep, produced across the resistor Rp, as a result of vibration shall be measured with a suitable device. This device shall have an appropriate voltage range, shall have the ability to measure, with an error of less than 10 percent, the rms value of a sine wave of voltage at all frequencies from 20 to 20,000 Hz. The value of the vibrational output Ep shall not exceed the limit specified herein at any point in the sweep frequency range during the last complete cycle of cycling vibration.

(C)15. Testing shall be performed approximately every 90 days, with sampling as follows:

n1 = 4	c1 = 0	
n2 = 4	c2 = 1;	where c2 represents the total allowable failures for the first and second samples combined.

In case of failure after double sampling, the failing test(s) shall become a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. After three consecutive successful submissions, the testing shall revert to the quarterly basis. None of the testing shall be considered as destructive except in case of failure.

16. Torque test shall be performed as follows:

    a. The tube shall be held securely at the cathode connection. A force of 5 pounds shall be applied to the heater cup without perceptible shock. This test may be made by applying the force at right angles to the inside of the cup at point 7/64 ±1/64 inch from the cathode end of the tube. An approved equivalent method may be used. The heater cup shall not loosen or short circuit on the cathode connection. This part of the test shall not be required if the space between the heater cup and the cathode sleeve is completely filled with insulating material.

    b. A torque of 15 inch-pounds shall be applied between anode and cathode without shock.

    c. A torque of 40 inch-pounds shall be applied between anode and grid without shock.

17. Test in jig made in accordance with Drawing 280-JAN. Each tube shall be subjected to a total of 15 shocks; that is, 5 shocks in each position Y, +Z, -Z in any sequency as shown on figure 1.

18. Voltage shall be 60 Hertz ac applied between anode and grid. No other voltages shall be applied. There shall be no evidence of failure as indicated by arc-over.

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

Agent:
DSA - ES

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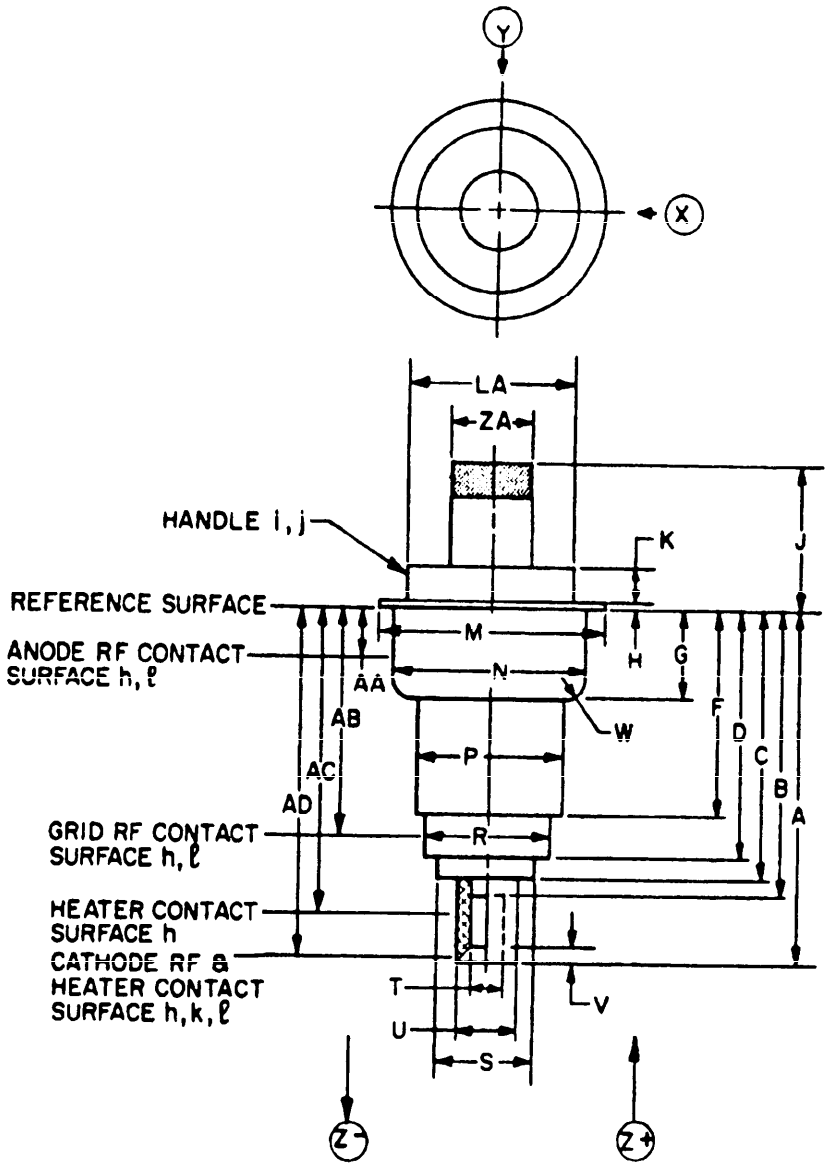


FIGURE 1. Outline drawing of electron tube type 7698.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	1.815(46.10)	1.875(47.62)	
B		1.534(38.96)	
C		1.475(37.46)	
D	1.289(32.74)	1.329(33.76)	
F	.970(24.64)	1.010(25.65)	
G	.462(11.73)	.477(12.12)	
J	.766(19.46)	.826(20.98)	
N	1.025(26.04)	1.035(26.29)	e, m
R	.655(16.64)	.665(16.89)	e, m
T	.213 (5.41)	.223 (5.66)	f, m
U	.315 (8.00)	.325 (8.26)	e, f, m
Quality conformance inspection, part 3			g
H		.040 (1.02)	
K		.185 (4.79)	
M	1.180(29.97)	1.195(30.35)	
P	.752(19.10)	.792(20.12)	
S		.545(13.84)	
V		.086 (2.18)	
W		.100 (2.54)	
LA	.840(21.34)	.860(21.84)	
ZA	.427(10.85)	.447(11.35)	
Electrode contact areas			n
AA	.035 (.89)	.361 (9.17)	a, e
AB	1.185(30.10)	1.265(32.13)	b, e
AC	1.534(38.96)	1.728(43.89)	c, f
AD	1.475(37.46)	1.815(46.10)	d, e, f

FIGURE 1. Outline drawing of electron tube type 7698 - Continued.

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NOTES:

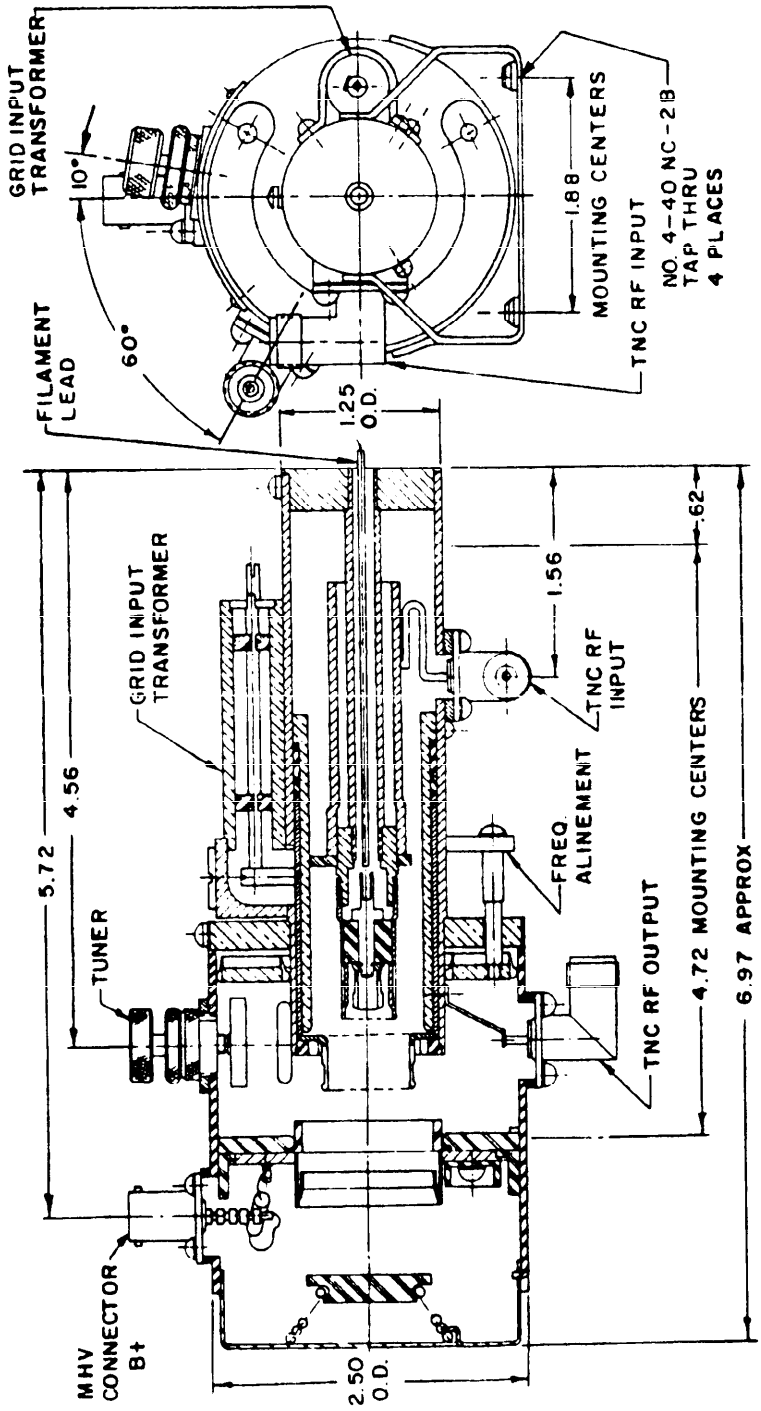
a. Anode rf contact surface and reference dimension for eccentricity measurements.
b. Grid rf contact surface and reference dimension for eccentricity measurements.
c. Heater contact surface and reference dimensions for eccentricity measurements.
d. Heater and cathode rf contact surface and reference dimensions for eccentricity measurements.
e. The total indicated runout of the anode and grid contact surface with respect to the cathode contact surface shall not exceed .020(.51 mm).
f. The total indicated runout of the cathode contact surface with respect to the heater contact surface shall not exceed .012(.30 mm).
g. These dimensions shall be tested on 10 tubes per month when in continuous productions. Failure of more than one tube to meet tolerances for any dimension shall cause that dimension to become, for all lots in process, part of quality conformance inspection, part 2.
h. Silver plated 30 MSI minimum.
i. Plating not required over radiator or handle support of copper, aluminum, or approved equivalent.
j. This surface shall be used for measurement of anode shank temperature.
k. Inner edge of heater and outer edge of cathode rf connection shall be free from burrs and sharp edges. Insulation material between heater and heater-cathode shall be securely affixed.
l. Total indicated runout (T.I.R.) of contact surfaces shall be gaged from center line of reference and shall be as follows. Note g shall apply.

Contact surface	T.I.R. (Maximum)	Reference
Anode	0.020	Cathode
Grid	0.020	Cathode
Heater	0.012	Cathode

m. Diameters N, R, T and U shall apply throughout entire contact areas as defined by dimensions AA, AB, AC, and AD respectively.
n. Dimensions in electrode contact areas table are for socket design purposes and are not intended for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 7698 - Continued.

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INCHES	MM
.62	15.75
1.25	31.75
1.56	39.62
1.88	47.75
2.50	63.50
4.56	115.82
4.72	119.89
5.72	145.29
6.97	177.04

NOTE: All dimensions are for reference only.

FIGURE 2. Pulse amplifier cavity.

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